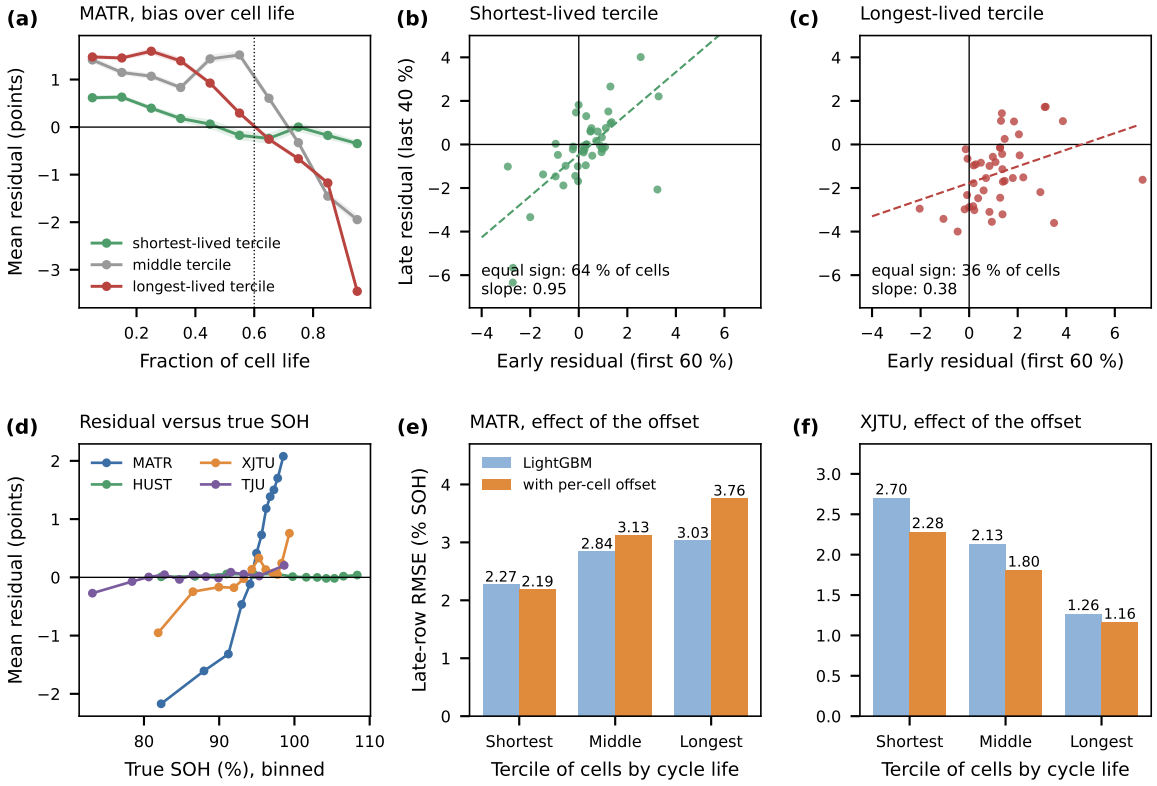
(a)
MATR, bias over cell life
Mean residual (points)
Fraction of cell life
shortest-lived tercile
middle tercile
longest-lived tercile
(b)
Shortest-lived tercile
Late residual (last 40 %)
Early residual (first 60 %)
equal sign: 64 % of cells
slope: 0.95
(c)
Longest-lived tercile
Early residual (first 60 %)
equal sign: 36 % of cells
slope: 0.38
(d)
Residual versus true SOH
Mean residual (points)
True SOH (%), binned
MATR
HUST
XJTU
TJU
(e)
MATR, effect of the offset
Late-row RMSE (% SOH)
LightGBM
with per-cell offset
2.27
2.19
2.84
3.13
3.03
3.76
Shortest
Middle
Longest
Tercile of cells by cycle life
(f)
XJTU, effect of the offset
2.70
2.28
2.13
1.80
1.26
1.16
Shortest
Middle
Longest
Tercile of cells by cycle life